Divyansh Shukla

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Education

Vellore Institute of Technology, Andhra Pradesh 2023 – 2027
B.Tech, Computer Science and Engineering (Artificial Intelligence & Machine Learning) *CGPA: 9.03 / 10*

DAV Public School, Sahibabad, *Class XII, CBSE* *2022* 91%

DAV Public School, Sahibabad, *Class X, CBSE* *2020* 95.6%

Technical Skills

Languages: Python, SQL, C++, JavaScript
Core CS: Data Structures and Algorithms, DBMS, Operating Systems, Computer Networks, Object-Oriented Programming, SDLC and Agile
Web & Backend: FastAPI, Django, REST APIs, PostgreSQL, HTML, CSS
Tools & Infrastructure: Git, Linux, Docker, Kubernetes (basics), AWS (EC2, S3, Lambda, IAM), Modal, Weights & Biases, Hydra
ML Frameworks: PyTorch, PyTorch Lightning, Hugging Face Transformers, Ultralytics (YOLO), Stable-Baselines3, NumPy, pandas, OpenCV
Deep Learning: object detection, flow and diffusion models, transformers, reinforcement learning (PPO)
Generative AI: Google Gen AI SDK (Gemini), hand-written agent loops (planning, memory, function calling), RAG with ChromaDB

Experience

Power Pulse India 2026 – Present
Machine Learning Intern *Ghaziabad, India*

- Trained a YOLOv8n thermal object detector on FLIR ADAS v2 across 6 road-safety classes. Cut missed pedestrians by 26% and raised mAP50-95 from 0.360 to 0.420, giving up 3 points of precision for the recall a safety case needs.
- Audited per-class crops and COCO size buckets to find the accuracy ceiling. 82% of pedestrian labels are under 32x32 px, and 37% of true trucks drew no prediction at all.
- Ran a four-experiment resolution and architecture sweep. Dropped a P2 detection head that scored below plain 640px at 1.51x the compute, and chose 800px over 960px.

Projects

RoPE-DiT: Diffusion Transformer with 2D Rotary Embeddings github.com/divyanshuklai/Custom2DRoPEDiT

- Wrote a 5M-parameter Diffusion Transformer from scratch in PyTorch: patchifier, timestep and class-label embedders, AdaLN-Zero conditioning, and a custom multi-head attention module.
- Built a 2D rotary positional embedding that rotates queries and keys by each patch's row and column, so attention sees relative 2D distance instead of a flat index.
- Trained on CIFAR-10 with a rectified flow objective. Sampled with a hand-written Euler solver and classifier-free guidance.

Fashion Search: Conversational Product Discovery github.com/divyanshuklai/fashion_search

- Built a FastAPI chat app that answers natural-language product queries across 10 Indian e-commerce stores, using Tavily search, Gemini 2.5 Flash extraction and BeautifulSoup scraping.
- Wrote the agent loop by hand on the Google Gen AI SDK, with no agent framework: a planner step decides whether to search or ask a clarifying question, conversation history acts as memory, and Gemini function calls are dispatched manually.
- Wrote a priority-based price and image extractor that tries JSON-LD product schema first, then Open Graph tags, then keyword-matched image tags.

Zero-Shot ICE RMA: Quadruped Locomotion RL github.com/divyanshuklai/zero-shot-ice-rma

- Built a MuJoCo environment for the Unitree Go1 with 30-dim proprioceptive observations, 12-dim joint-position actions and fractal terrain, then debugged a PPO baseline that ran 42M steps without learning a gait and traced the exploding value loss to unnormalized observations.

Certifications & Achievements

NPTEL: Deep Learning for Computer Vision (Topper, top 1% of 463, IIT-H); Applied Accelerated AI (Topper, top 2% of 591, IIT-G)
Coursera: Deep Learning Specialization | **AWS**: AWS Academy Cloud Architecting, AWS Cloud Practitioner (badge)
Achievement: Won the intra-campus round of the Smart India Hackathon at VIT-AP, 2024